

Isaiah Foster

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EDUCATION

Washington State University

B.S. Computer Science, GPA: 4.0/4.0

Center of Academic Excellence in Cyber Operations Certificate

VICEROY Scholar (DoD-funded cyber operations program)

Courses: Algorithms, Systems Programming, Programming Language Design

Pullman, WA

Aug. 2024 – May 2028 (Expected)

Alongside degree

Aug. 2024 – Present

EXPERIENCE

Linux Software Engineering Intern

Mar. 2026 – Present

Schweitzer Engineering Laboratories

Pullman, WA

- Refactored a C++ firewall daemon for proprietary network switches to apply rule updates atomically, eliminating a timing gap where all traffic was dropped; wrote 15 functional tests and 3 unit tests covering rule application
- Restored a Jenkins pipeline of 1,700+ tests that had been broken for 3 months by refactoring a configuration with Groovy, and revived an abandoned C# test suite by fixing over 200 failing unit tests required to ship regulation-compliant firmware images
- Built a cryptographic dev tool to generate PKCS#12 and X.509 test artifacts for on-device certificate import/export, including deliberately malformed files for negative testing of parsing and validation logic
- Wrote Python functional tests across a large firmware monorepo, debugging failures through the full stack: on-device behavior, Yocto/BitBake builds, C++ daemons, and C# orchestration; reviewed PRs for security-critical C++ code

Teaching Assistant, C/C++ Data Structures

Jan. 2026 – May 2026

Washington State University

Pullman, WA

- Led a lab section of 17 students weekly; taught foundational concepts and helped students debug code

Cybersecurity Intern

Jun. 2025 – Aug. 2025

Air Force Research Lab/Griffiss Institute

Rome, NY

- Performed vulnerability analysis on a proprietary operating system using WolfSSL, revealing issues missed in a six-figure third-party report
- Placed 1st out of 15 teams in the final intern cyber operation, using iptables, netcat, Ghidra, and netmap
- Presented findings to Pentagon cyber professionals; work recognized with the New York State Assembly Certificate of Merit

Undergraduate Researcher, LLM Privacy & Security

Jan. 2025 – May 2025

Washington State University, School of EECS

github.com/isaiah-foster/LLM-defense-research-project

- Built an attack framework in Python to test LLMs for PII leakage, implementing extraction, reconstruction, and inference attacks against fine-tuned GPT-2 and OpenAI API models; used Faker-generated synthetic PII and Microsoft Presidio to measure leakage
- Implemented Proactive Privacy Amnesia in PyTorch and Hugging Face Transformers, a two-phase fine-tuning defense that unlearns memorized PII, cutting leakage by 55% at a cost of 5% higher perplexity
- Presented findings as a poster at CySER 2025 under mentorship of a PhD student and faculty advisor

PROJECTS

VisualNetwork | C, Go, eBPF (XDP/TC), Libbpf, Linux, Make

github.com/isaiah-foster/Visual-Network

- Performed all kernel-side development of a Linux network traffic analyzer on a small Agile team, owning the eBPF layer and its interface to userspace
- Wrote C eBPF programs attached to XDP (ingress) and TC (egress) hooks to classify inbound and outbound packets by protocol and address, maintaining counters for each in BPF maps
- Built a Go userspace daemon that reads kernel maps via a ring buffer, aggregates volume, protocol, and latency metrics, and serves them to a React dashboard through a reverse proxy

SKILLS & ACTIVITIES

Languages: C++, Python, C, C#, Go, Bash

Systems & Networking: Linux, iptables, Wireshark, eBPF, tcpdump, SSH, TCP/IP, HTTP

Tools: GoogleTest, Git, Jenkins, Yocto, BitBake, Gherkin, Behave, Docker

Leadership: Vice President and founding member, Triangle Fraternity (WSU's STEM fraternity)